

CST8246 - Network Setup and Automated Network Configuration Verification

Lab Overview

- Set up a functional networking environment within the lab.
- Configure and manage network settings on virtualized Linux systems.
- Automate key networking tasks using shell scripting.

Lab Outcomes

- Implement both dynamic (DHCP) and static IP addressing.
- Establish proper communication between the Linux server and client.
- Develop a script to automate network configuration verification.

Lab Deliverables

- A fully functional Linux-based server and client with correct network configurations.
- An automation script to validate network settings.
- Full demonstration requirements are available in Brightspace, where you downloaded the lab.

Section A – Network Configuration

1. Pre-configuration Tasks

1.1 VMware Network Configuration

- Open **VMware Virtual Network Editor**.
- Ensure the following settings:
 - One **VMnet** is configured in **NAT mode** to allow the virtual machines to access the Internet via the host system's wireless adapter.
 - Another **VMnet** is configured in **Bridged mode**, linking it to the physical network interface. This adapter must be connected to a **segregated network (RED network)**.

1.2 Verifying VMware Machines Settings

- Confirm that each virtual machine has **two Ethernet adapters**.

- If any adapter is missing, add the required **VMnet connections**.
- Assign the adapters as follows:
 - **Primary adapter (Blue network)** – Provides external Internet access.
 - **Secondary adapter (Red network)** – Connects to the isolated internal network.

2. Guest Operating System Configuration

2.1 Security Configurations

- **Firewall Settings:** Disable the firewall to avoid connectivity issues during the lab setup.
- **SELinux Configuration:**
 - **Default:** Keep SELinux enabled for security purposes.
 - **Alternative:** If experiencing difficulties, SELinux can be disabled.
 - If you encounter any SELinux-related problems, consult your lab professor for guidance.

2.2 User Account Creation

- Create a dedicated user account for testing services within the lab environment.
- Use the following credentials:
 - **Username:** cst8246
 - **Password:** cst8246
- Grant administrative privileges:
 - Add the user to the "wheel" group for sudo access.
 - If necessary, use the following command:

```
sudo usermod -aG wheel cst8246
```

2.3 Additional Notes

- It is recommended to create this user with administrative privileges during the **Red Hat installation process**.
- Ensure that all security configurations align with industry best practices.

Section B – Network Service Management

1. Ensuring NetworkManager is Active

- Make sure the NetworkManager service is running and enabled:

- `systemctl restart NetworkManager`

Section C – Network Configuration in Linux

1. General Network Configuration

- Network interfaces must be configured with IP information, either manually using **ip addr** or via the network service and configuration files.
- IP parameters for the network service are provided in configuration files located at:

- `/etc/sysconfig/network-scripts/ifcfg-interface`

- Routing information (the routing table) can be viewed using:

- `ip route`

- Hostname resolution can be performed via `/etc/hosts` or DNS.
- DNS servers used for queries are listed in:

- `/etc/resolv.conf`

- The file **/etc/nsswitch.conf** lists the name resolution mechanisms in priority order.
- Hostname viewing & setting is accomplished with the `hostnamectl` command.

2. Specific Network Configuration Commands

- List network interfaces detected by the kernel:

- `lspci` (two Ethernet adapters should be listed)

- View the current network setup:
 - `ip addr show [device]`
 - first interface (device name for “eth0”)
 - second interface (device name for “eth1”)
 - are any IP addresses listed?
- Check routing table:
 - `ip route show`
- View hostname settings:
 - `hostnamectl`
- Check current DNS servers:
 - `cat /etc/resolv.conf`
- Identify hostname resolution order:
 - `cat /etc/nsswitch.conf` (look under the “hosts” entry).

Section D – Network Configuration Using DHCP

1. Configuring DHCP for Blue Network

1. Edit the interface configuration file for the first interface (eth0, ens160, or equivalent):
2. If the configuration file does not exist, create it at:
 - `/etc/sysconfig/network-scripts/ifcfg-interface`
3. Remember that commenting out items reverts them to their default values.
4. Do you know what the default value is?

5. To activate the device, set `ONBOOT=yes`.
6. To disable IPv6, set IPv6 directives to `no`.
7. These files will be modified in future labs.

Your network configuration (with interfaces like `ens160` and `ens224`) should look something like this:

```
/etc/sysconfig/network-scripts/ifcfg-ens160:
##### BLUE NETWORK FOR BOTH SERVER AND CLIENT
TYPE=Ethernet
PROXY_METHOD=none
BROWSER_ONLY=no
BOOTPROTO=dhcp
DEFROUTE=yes
IPV4_FAILURE_FATAL=no
IPV6INIT=yes
IPV6_AUTOCONF=yes
IPV6_DEFROUTE=yes
IPV6_FAILURE_FATAL=no
IPV6_ADDR_GEN_MODE=eui64
NAME=ens160
UUID=3a21f904-ad9a-4639-b192-6eab8904f0fa
DEVICE=ens160
ONBOOT=yes
```

```
/etc/sysconfig/network-scripts/ifcfg-ens224:
##### RED NETWORK CLIENT
TYPE=Ethernet
PROXY_METHOD=none
BROWSER_ONLY=no
BOOTPROTO=none
IPADDR=172.16.31.167
NETMASK=255.255.0.0
NETWORK=172.16.0.0
BROADCAST=172.16.255.255
DEFROUTE=yes
IPV4_FAILURE_FATAL=no
IPV6INIT=yes
IPV6_AUTOCONF=yes
IPV6_DEFROUTE=yes
IPV6_FAILURE_FATAL=no
IPV6_ADDR_GEN_MODE=eui64
NAME=ens224
DEVICE=ens224
ONBOOT=yes
```

```
/etc/sysconfig/network-scripts/ifcfg-ens224:
##### RED NETWORK SERVER
TYPE=Ethernet
PROXY_METHOD=none
BROWSER_ONLY=no
BOOTPROTO=none
IPADDR=172.16.30.167
NETMASK=255.255.0.0
NETWORK=172.16.0.0
BROADCAST=172.16.255.255
DEFROUTE=yes
IPV4_FAILURE_FATAL=no
IPV6INIT=yes
IPV6_AUTOCONF=yes
IPV6_DEFROUTE=yes
IPV6_FAILURE_FATAL=no
IPV6_ADDR_GEN_MODE=eui64
NAME=ens224
DEVICE=ens224
ONBOOT=yes
```

2. Network Startup and Interface Verification

- View log file:
 - `tail -f /var/log/messages`
 - Note: Keep the log file running throughout the lab. It's helpful to keep the window wide and positioned at the bottom of your screen.

- Start or restart NetworkManager:
 - `systemctl restart NetworkManager`
 - Verify interface configuration:
 - `ip addr show`
-

Section E – Network Configuration Using Static IPs

1. Configuring Static IP for RED Network (closed network)

Edit your interface configuration file to assign a static IP address:

- **For the server:** Use 172.16.30.<your_magic_number>, ensuring all essential parameters are set, including the netmask (255.255.0.0). **Do not** configure a gateway, as the RED network is isolated.
 - **Example:** If your magic number is **101**, the server IP address will be **172.16.30.101**.
 - *Note:* Your magic number is available under "Content" on Brightspace.
- **For the client:** Use 172.16.31.<your_magic_number>, with the same netmask (255.255.0.0). **Do not** configure a gateway.
 - **Example:** If your magic number is **101**, the client IP address will be **172.16.31.101**.
- *Note:* Interface configuration parameters are **case-sensitive**.
- **Activate the device:** Set ONBOOT=yes.
- **Disable IPv6.**

2. Routing Verification

- Identify the default gateway:
 - `ip route show`
- *Note:* The DHCP server provides the gateway information. The DEFROUTE option determines whether to accept the DHCP-assigned gateway as the default. If set to "no," there will be no gateway, no Internet, and no access to Google...!

Give it a try: check the error message your browser shows (could be a test question?).

3. Resolver Configuration

- Task: Determine the IP addresses of your name server(s) from the `/etc/resolv.conf` file.
- Note: This information comes from the DHCP server. The PEERDNS option controls whether to accept the DNS servers assigned by the DHCP server. If set to "no," there will be no name resolution, no Internet, and no access to Google...!
- Try it out: Record the error message displayed by your browser (this might be a test question).
- Question: When configuring our own DNS service on a server (lab coming soon), should we change the default value of the PEERDNS option?
- Identify the search domain(s).

Section F – Testing Network Configuration

Verify and Test Your Interface Configuration Using CLI Utilities

- Check Interface Configuration:
 - View MAC address
 - Check IP address
 - Identify TCP/IP configuration (static or dynamic)
- Examine the Routing Table
 - Perform Ping Tests:
 - Ping your own IP address
 - Ping the server from a client using its IP address

- Ping one or both name servers
- Test Internet Connectivity:
 - Browse the web to confirm external access
 - Note: If your blue/college network is not working after setup, try reversing the configuration in either the sysconfig files or VMware settings for your virtual machine.
ping [your server IP]

Section G – Hostname Setting & Resolution

1. Setting Hostname

- Change hostname using `hostnamectl`:
 - `hostnamectl set-hostname [your_network_id]-SRV.exampleMN.lab`
- The system type is used to distinguish between your client and your server (SRV/CLT). MN stands for your magic number. Example: smit0001-SRV.example99.lab, smit0001-CLT.example99.lab

2. Testing Hostname Resolution

- Include your hostname and static IP address in the hosts file.
- *Question:* Should the “localhost” entry be modified, removed, or kept?

3. Hostname Resolution Using Name Servers

- Check the file that contains the name server entries once more.

4. Test Hostname Resolution

- Use the hostname to ping your own system.
 - Note: Should this work?

- From a client, use the hostname to ping your server.
 - Note: Should this work? Why or why not? Does it depend on your configuration?

Section H – Summary of Network Information

Summary of the Server System

Search item	File/command	Output
Hostname		
Hostname resolution order		
DNS server(s)		
“blue interface”: Default gateway		
“blue interface”: IP address & MAC address		
“red interface”: IP address & MAC address		

Final Testing

- Test again: After completing the network setup, reboot and verify your interfaces.
- Create a backup or snapshot.

Section I – Scripting

- Create a script to summarize network settings and display results
 - **Script Name:** testnet.sh

- **Required Header Information:** Include details such as purpose, date, student name, and student number.
- **Documentation:** Provide brief comments throughout the script.
- **Output Requirements:** The script should present only the required details in three clearly separated sections. The output must be concise, well-structured, and fit within a single screen.
- Verify service status:
 - Ensure the **NetworkManager** service is active.
 - Confirm that **Firewalld** is disabled and **iptables** is turned off (or flushed).
 - Verify that **SELinux** is set to the desired status.
- Display Network Configuration:
 - Show the **IP addresses** for both the "**red**" and "**blue**" interfaces, along with their names.
 - List the configured **DNS servers**.
 - Display the **hostname resolution order**.
 - Show the contents of the **hosts file**.
 - Identify the **default gateway**.
- Perform connectivity tests:
 - **Ping Test 1:** Verify connectivity between the **server** and the **client** (red network).
 - **Ping Test 2:** Check connectivity between the **server** and the **default gateway** (blue network).

- **Final Step:** Run and test the script.